

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2018-2019)

PRE-MEDICAL : NURTURE COURSE [PHASE-I, II & III]

TEST # 15

DATE : 03 - 02 - 2019

Test Type : Minor (CBT)

Test Pattern : AIIMS

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **180**.
5. **Each correct answer carry 1 mark. While one third mark will be deducted for each incorrect answer.**
6. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
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1. A gas bubble oscillates with a time period T proportional to $P^a d^b E^c$ where P is pressure, d is the density and E is the energy. The values of a , b & c are.

(1) $a = \frac{3}{2}, b = -\frac{1}{3}, c = \frac{1}{2}$

(2) $a = -\frac{5}{6}, b = \frac{1}{3}, c = \frac{1}{2}$

(3) $a = -\frac{5}{6}, b = \frac{1}{2}, c = \frac{1}{3}$

(4) $a = \frac{3}{2}, b = -\frac{1}{3}, c = \frac{1}{2}$

2. If vectors $\vec{a} = 3\hat{i} + 8\hat{j} - 2\hat{k}$ and

$\vec{b} = 6\hat{i} + 16\hat{j} + x\hat{k}$ are parallel to each other, then the value of x will be :

- (1) +8 (2) +4 (3) -4 (4) -8

3. A particle moves in a straight line so that $s = \sqrt{t}$, then its acceleration is proportional to :

(1) (Velocity) (2) (Velocity)^{3/2}

(3) (Velocity)³ (4) (Velocity)²

4. A physical quantity P is given by

$P = \frac{A^2 \sqrt{B}}{C}$, and the percentage errors in

the measurements of A , B and C are 1%, 2% and 4% respectively. Find the percentage error in P .

- (1) 8 (2) 6 (3) 7 (4) 2

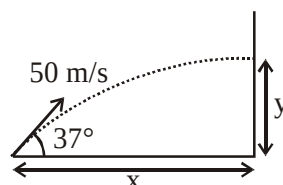
5. A particle moves with an acceleration $a = k\sqrt{t}$. If initial velocity of particle is zero then velocity of particle as a function of time t is :

(1) $\frac{2}{3}kt^{3/2}$ (2) $\frac{2}{3}kt^{1/3}$

(3) $\frac{1}{3}kt^{1/2}$ (4) $\frac{2}{3}kt$

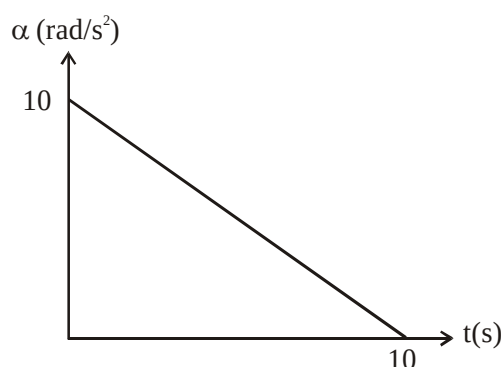
6. A particle is projected as shown in figure. If the particle strikes at the wall after 1 sec,

then the ratio $\frac{x}{y}$ will be (approximately)



- (1) 1 (2) 1.6
(3) 2.6 (4) 3

7. A body rotating about an axis has an angular acceleration, which varies with time, according to graph shown



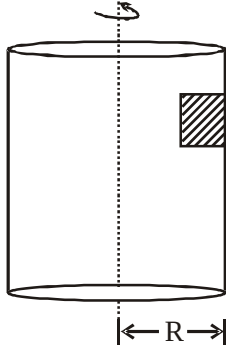
If initial angular velocity of body is zero, then its angular velocity at the end of 5 s is :-

- (1) 12.5 rad/s
(2) 25 rad/s
(3) 37.5 rad/s
(4) 50 rad/s

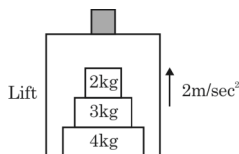
8. A block of mass 1 kg slides on a rough horizontal surface. If the speed of the block decreases from 10 m/s to 8 m/s, the thermal energy developed in this process :-

- (1) 50 J (2) 18 J
(3) 36 J (4) 25 J

9. A block is rotating in contact with vertical wall (rotor) as shown. Find the minimum value of ω so that the block does not slide down.



- (1) $\sqrt{\frac{2g}{\mu R}}$ (2) $\sqrt{\frac{3g}{\mu R}}$
(3) $\sqrt{\frac{g}{\mu R}}$ (4) $\sqrt{\frac{4g}{\mu R}}$
10. A body radiates energy 5W at a temperature of 127°C. If the temperature is increased to 927°C, then it radiates energy at the rate of—
(1) 410 W (2) 81 W
(3) 405 W (4) 20 W
11. A lift is moving upwards with an acceleration of 2 m/sec². Inside the lift a 4 kg block is kept on the floor. On the top of it, 3 kg block is placed and again a 2 kg block is kept on the 3 kg block. Calculate contact force between 2 kg block and the 3 kg block.



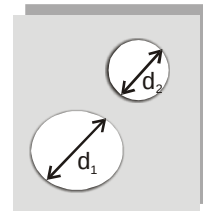
- (1) 24 N (2) 48 N
(3) 12 N (4) 36 N
12. Liquids A and B are at 30°C and 20°C. When mixed in equal masses, the temperature of the mixture is found to be 26°C. Their specific heat are in the ratio of :-
(1) 3 : 2 (2) 1 : 1
(3) 2 : 3 (4) 4 : 3

13. A particle starts from rest moving along x-axis, under the influence of a force

$$F = 3x^2 + 2x$$

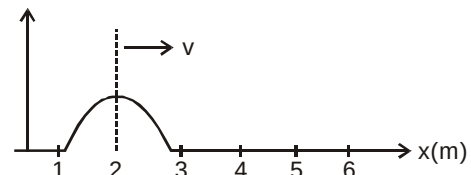
When particle moves from $x = 1$ to $x = 2$ m its velocity changes to 2 m/s. Mass of particle is :-

- (1) 5 kg (2) 2 kg
(3) 1 kg (4) $\frac{1}{2}$ kg
14. Two holes of unequal diameters d_1 and d_2 ($d_1 > d_2$) are cut in a metal sheet.

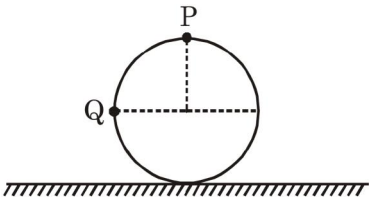


If the sheet is heated

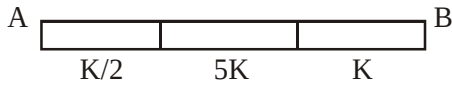
- (1) both d_1 and d_2 will decrease
(2) both d_1 and d_2 will increase
(3) d_1 will increase, d_2 will decrease
(4) d_1 will decrease, d_2 will increase
15. Wave pulse on a string shown in figure is moving to the right without changing shape. Consider two particles at position $x_1 = 1.5$ m and $x_2 = 2.5$ m. Their transverse velocities at the moment shown in figure are along directions



- (1) positive y-axis and positive y-axis, respectively
(2) negative y-axis and positive y-axis, respectively
(3) positive y-axis and negative y-axis, respectively
(4) negative y-axis and negative y-axis, respectively

16. A liquid with coefficient of volume expansion γ is filled in a container of a material having the coefficient of linear expansion α . If the liquid overflows on heating, then :-
- $\gamma = 3\alpha$
 - $\gamma > 3\alpha$
 - $\gamma < 3\alpha$
 - $\gamma = 3\alpha^3$
17. The phase difference between two SHM $Y_1 = 10\sin(10\pi t + \pi/3)$ and $Y_2 = 12\sin(8\pi t + \pi/4)$ at $t = 0.5s$ is
- $\frac{11\pi}{12}$
 - $\frac{13\pi}{12}$
 - π
 - $\frac{17\pi}{12}$
18. A pendulum clock keeps correct time at 0°C . The thermal coefficient of linear expansion of the material of the pendulum is α . If the temperature rises to $t^\circ\text{C}$ then the clock loses per day by :-
- $\alpha t \text{ sec}$
 - $\frac{1}{2}\alpha t \text{ sec}$
 - $\alpha t \times 86400 \text{ sec}$
 - $\frac{1}{2}\alpha t \times 86400 \text{ sec}$
19. Calculate K.E. of rotation of a circular disc of mass 1 kg and radius 0.2 m rotating about an axis passing through its center and perpendicular to its plane. The disc is making $30/\pi \text{ rpm}$.
- 0.01 joule
 - 0.04 joule
 - 0.06 joule
 - 0.08 joule
20. A ball is rolling on the ground. The ratio of speed at P and speed at Q is :
- 
- $\sqrt{2} : 1$
 - $2 : 1$
 - $2\sqrt{2} : 1$
 - $1 : 2$
21. A body is rolling without slipping on a horizontal surface and its rotational kinetic energy is equal to the translational kinetic energy. The body is
- disc
 - sphere
 - cylinder
 - ring
22. When a sphere rolls without slipping, the ratio of its kinetic energy of translation to its total kinetic energy is
- $1 : 7$
 - $1 : 2$
 - $1 : 1$
 - $5 : 7$
23. A particle is executing SHM with amplitude A and has maximum velocity V_0 . Its speed at displacement $A/2$ will be
- $(\sqrt{3})V_0/2$
 - $V_0/\sqrt{2}$
 - V_0
 - $V_0/4$
24. The KE and PE of a particle executing SHM with amplitude A will be equal when its displacement is
- $A\sqrt{2}$
 - $A/2$
 - $A/\sqrt{2}$
 - $A\sqrt{2/3}$

25. AA composite rod made of 3 rods of equal length and cross-section is shown in figure. The end A and B are at constant temperatures. The effective thermal conductivity of the bar is :-

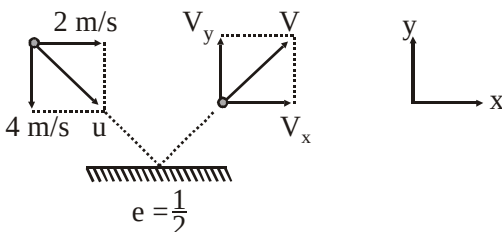


- (1) 15 K/16
(2) 6 K/13
(3) 5 K/16
(4) 2 K/13
26. Particles of masses $m, 2m, 3m, \dots, nm$ are placed on the same line at distances $L, 2L, 3L, \dots, nL$ from O. The distance of center of mass from O is

- (1) $\frac{(2n+1)L}{4}$
(2) $\frac{L}{(2n+1)}$
(3) $\frac{n(n^2+1)L}{2}$
(4) $\frac{(2n+1)L}{3}$

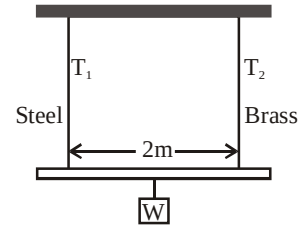
27. In the figure shown, a small ball hits obliquely a smooth and horizontal surface with speed u whose x and y components are indicated. If the coefficient

of restitution is $\frac{1}{2}$, then its x and y components v_x and v_y just after collision are respectively :-



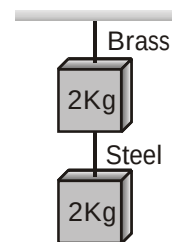
- (1) 4 m/s, 1 m/s
(2) 2 m/s, 1 m/s
(3) 2 m/s, 2 m/s
(4) 4 m/s, 2 m/s

28. A light rod of length 2m suspended from the ceiling horizontally by means of two vertical wires of equal length. A weight W is hung from a light rod as shown in figure. The rod hung by means of a steel wire of cross-sectional area $A_1 = 0.1 \text{ cm}^2$ and brass wire of cross-sectional area $A_2 = 0.2 \text{ cm}^2$. To have equal stress in both wires, $T_1/T_2 =$

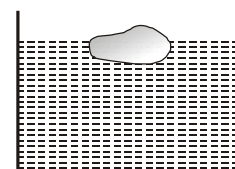


- (1) $1/3$ (2) $1/4$ (3) $4/3$ (4) $1/2$
29. If the ratio of lengths, radii and Young's modulus of steel and brass wires shown in the figure are a, b and c , respectively. The ratio between the increase in lengths of steel and brass wires would be :-

- (1) $\frac{b^2 a}{2c}$
(2) $\frac{bc}{2a^2}$
(3) $\frac{ba^2}{2c}$
(4) $\frac{a}{2b^2 c}$

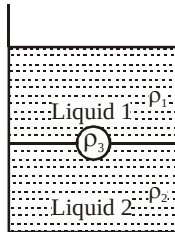


30. A body floats in a liquid contained in a beaker. The whole system as shown falls freely under gravity. The upthrust on the body due to the liquid is :-

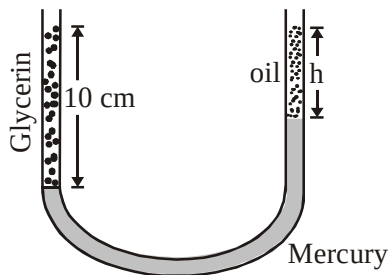


- (1) Zero
(2) Equal to the weight of the liquid displaced
(3) Equal to the weight of the body in air
(4) Equal to the weight of the immersed portion of the body

31. Diagram shows a jar filled with two non mixing liquids 1 and 2 having densities ρ_1 and ρ_2 respectively. A solid ball, made of a material of density ρ_3 , is dropped in the jar. It comes to equilibrium in the position shown in the figure. Which of the following is true for ρ_1 , ρ_2 and ρ_3 ?

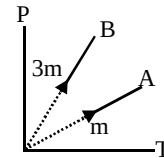


- (1) $\rho_3 < \rho_1 < \rho_2$
 (2) $\rho_1 < \rho_3 < \rho_2$
 (3) $\rho_1 < \rho_2 < \rho_3$
 (4) $\rho_1 > \rho_3 > \rho_2$
32. A vertical U-tube of uniform inner cross section contains mercury in both sides of its arms. A glycerin (density = 1.3 g/cm^3) column of length 10 cm is introduced into one of its arms. Oil of density 0.8 g/cm^3 is poured into the other arm until the upper surfaces of the oil and glycerin are in the same horizontal level. Find the length of the oil column, (Density of mercury = 13.6 g/cm^3) :-

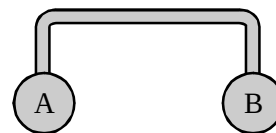


- (1) 10.4 cm (2) 8.2 cm
 (3) 7.2 cm (4) 9.6 cm

33. Two different masses m and $3m$ of an ideal gas are heated separately in a vessel of constant volume. The graph between the pressure P and absolute temperature T for these two cases are shown in the figure as A and B. The ratio of slopes of curves B to A is :-

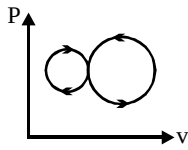


- (1) 3 : 1 (2) 1 : 3
 (3) 9 : 1 (4) 1 : 9
34. Considering the gases to be ideal, the value of $\gamma = \frac{C_p}{C_v}$ for a gaseous mixture consisting of 3 moles of carbon dioxide and 2 moles of oxygen will be ($\gamma_{O_2} = 1.4, \gamma_{CO_2} = 1.3$) :-
- (1) 1.37 (2) 1.33
 (3) 1.55 (4) 1.63
35. Two spherical vessel of equal volume, are connected by a narrow tube. The apparatus contains an ideal gas at one atmosphere and 300 K. Now if one vessel is immersed in a bath of constant temperature 600 K and the other in a bath of constant temperature 300 K. Then the common pressure will be :-



- (1) 1 atm (2) $\frac{4}{5}$ atm
 (3) $\frac{4}{3}$ atm (4) $\frac{3}{4}$ atm

36. In the following indicator diagram, the net amount of work done will be :-



- (1) Positive
(2) Negative
(3) Zero
(4) infinity
37. When an ideal diatomic gas is heated at constant pressure, the fraction of the heat energy supplied which increases the internal energy of the gas is :-

(1) $\frac{2}{5}$

(2) $\frac{3}{5}$

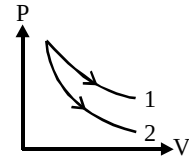
(3) $\frac{3}{7}$

(4) $\frac{5}{7}$

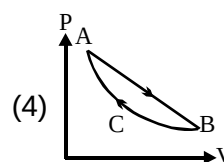
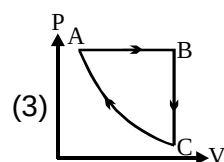
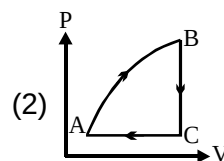
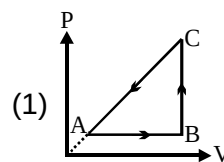
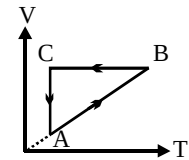
38. Initial pressure and volume of a gas are P and V respectively. First it is expanded isothermally to volume $4V$ and then compressed adiabatically to volume V . The final pressure of gas will be (given $\gamma = 3/2$) :-

- (1) $1P$
(2) $2P$
(3) $4P$
(4) $8P$

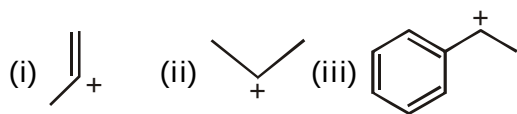
39. P - V plots for two gases during adiabatic process are shown in the figure. Plots 1 and 2 should correspond respectively to :-



- (1) He and O_2
(2) O_2 and He
(3) He and Ar
(4) O_2 and N_2
40. A cyclic process ABCA is shown in the V - T diagram. Process on the P - V diagram is :-

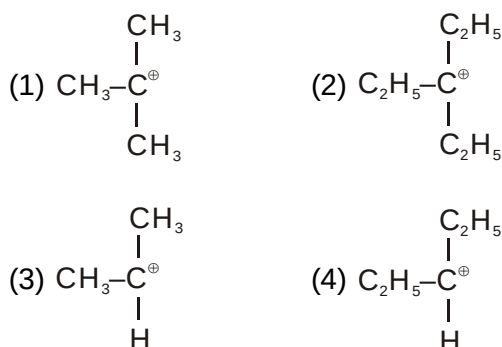


41. The relative stability of the following carbocations is :-

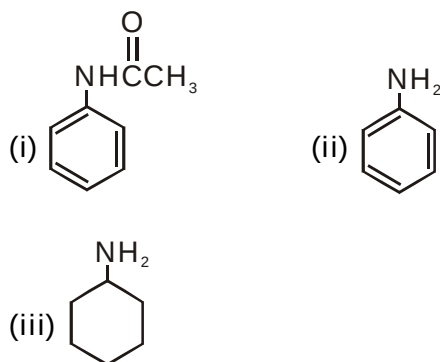


- (1) i > ii > iii
(2) ii > iii > i
(3) i > iii > ii
(4) iii > ii > i

42. Which of the following carbonium ion is most stable?

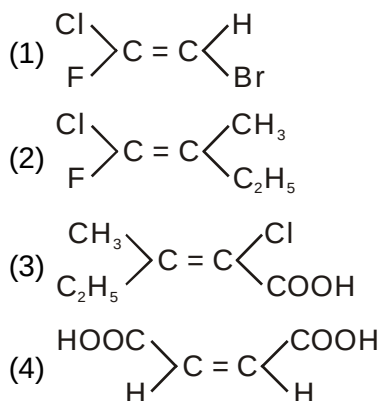


43. Rank of the following three compounds in decreasing order of basicity is :-

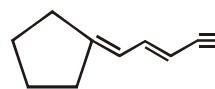


- (1) iii > i > ii (2) iii > ii > i
(3) ii > i > iii (4) ii > iii > i

44. The "Z"-isomer is :-

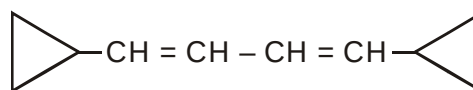


45. The number of cis-trans isomer possible for the following compound is :-



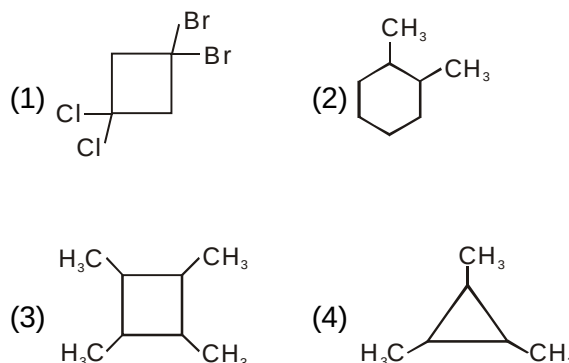
- (1) 2 (2) 4
(3) 6 (4) 8

46. The total number of geometrical isomers possible in following compound is :-

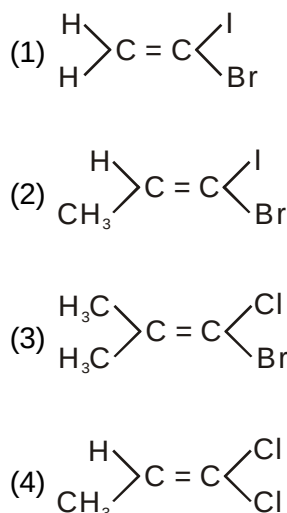


- (1) 4 (2) 6 (3) 3 (4) 2

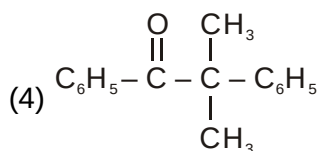
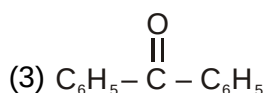
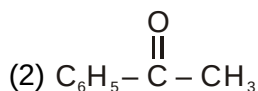
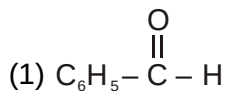
47. Which of the following compounds cannot show geometrical isomerism?



48. Geometrical isomerism is shown by :-

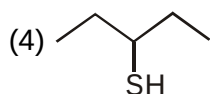
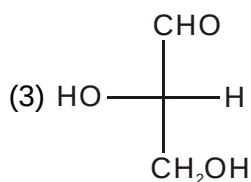
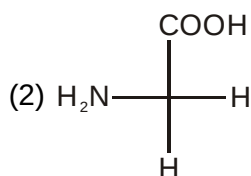


49. Keto-enol tautomerism is observed in :-

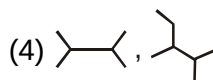
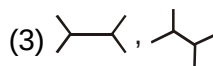
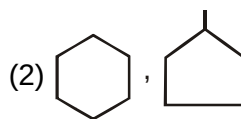
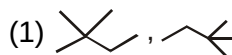


50. Which of the following molecules is expected to rotate the plane of polarised light?

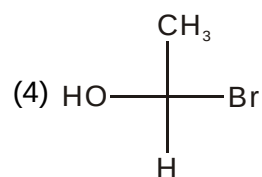
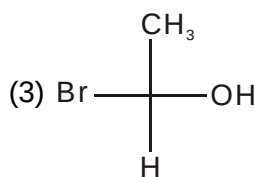
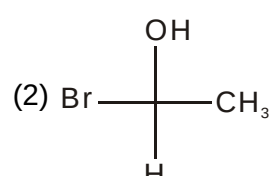
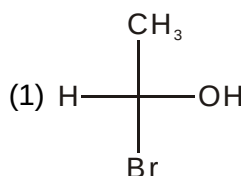
(1) Methane



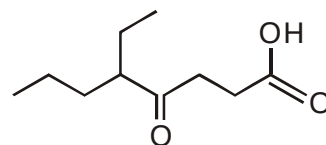
51. Which of the following pair is the chain isomer?



52. Which of the following compound has "S" configuration?

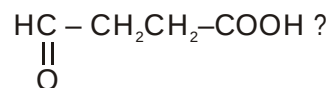


53. The IUPAC name of the compound shown here is :-



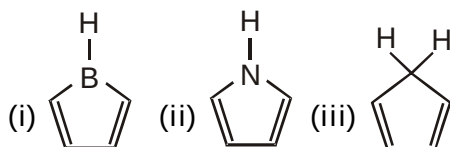
- (1) 5-Ethyl-1-hydroxyoctan-1,4-dioic acid
(2) 2-Carboxyethyl-3-hexyl ketone
(3) 3-(3-Hexylcarbonyl) propanoic acid
(4) 5-Ethyl-4-ketooctanoic acid

54. What is the ratio of number of σ - and π -bonds in the molecule



- (1) 5 (2) 6 (3) 4 (4) 8

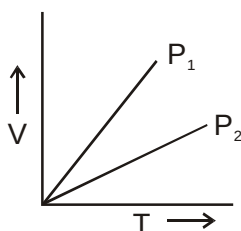
55. Choose the following species that would be predicted to be aromatic according to Hückel's rule.



- (1) i (2) ii
(3) iii (4) i and ii
56. Transition from $n = 4, 5, 6$ to $n = 3$ in hydrogen spectrum gives :-
(1) Lyman series
(2) Paschen series
(3) Balmer series
(4) Pfund series
57. Which transition of electron in the hydrogen atom emits maximum energy?
(1) $2 \rightarrow 1$ (2) $1 \rightarrow 4$
(3) $4 \rightarrow 3$ (4) $3 \rightarrow 2$
58. Which one represents an impossible arrangement?
- | n | l | m | s |
|-------|-----|-----|-------|
| (1) 3 | 2 | -2 | $1/2$ |
| (2) 4 | 0 | 0 | $1/2$ |
| (3) 3 | 2 | -3 | $1/2$ |
| (4) 5 | 3 | 0 | $1/2$ |
59. The number of radial nodes of 3s and 3p-orbitals are respectively :-
(1) 2, 0 (2) 0, 2
(3) 1, 2 (4) 2, 1
60. Which is not a state function?
(1) Internal energy
(2) Entropy
(3) Work
(4) Enthalpy
61. In which reaction ΔS is positive?
(1) $\text{H}_2\text{O}_{(l)} \longrightarrow \text{H}_2\text{O}_{(s)}$
(2) $3\text{O}_{2(g)} \longrightarrow \text{H}_2\text{O}_{(g)}$
(3) $\text{H}_2\text{O}_{(l)} \longrightarrow \text{H}_2\text{O}_{(g)}$
(4) $\text{N}_{2(g)} + 3\text{H}_{2(g)} \longrightarrow 2\text{NH}_{3(g)}$

62. The bond energy of C – H, C – C, C = C and H – H bonds are 414, 347, 615 and 435 kJ mol⁻¹ at 298 K respectively. The enthalpy change for the reaction;
 $\text{CH}_2 = \text{CH}_{2(g)} + \text{H}_{2(g)} \longrightarrow \text{CH}_3 - \text{CH}_{3(g)}$
would be :-
(1) +125 kJ
(2) -125 kJ
(3) +250 kJ
(4) -250 kJ
63. For the reaction, $\text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI}$:-
(1) $K_c = 2K_p$
(2) $K_c > K_p$
(3) $K_c = K_p$
(4) $K_c < K_p$
64. When KOH is dissolved in water, heat is evolved. If the temperature is raised, the solubility of KOH :-
(1) Increases
(2) Decreases
(3) Remains the same
(4) Cannot be predicted
65. For which salt the pH of its solution does not change with dilution?
(1) NH_4Cl
(2) $\text{CH}_3\text{COONH}_4$
(3) CH_3COONa
(4) None of these
66. Which one is amphoteric oxide?
(1) SO_2 (2) B_2O_3
(3) ZnO (4) Na_2O
67. Let the solubilities of AgCl in H_2O , 0.01 M CaCl_2 , 0.01M NaCl and 0.05M AgNO_3 be s_1 , s_2 , s_3 and s_4 respectively. Then correct relationship between these quantities?
(1) $s_1 > s_2 > s_3 > s_4$
(2) $s_1 > s_2 = s_3 > s_4$
(3) $s_1 > s_3 > s_2 > s_4$
(4) $s_4 > s_2 > s_3 > s_1$

68. V versus T curves at constant pressure P_1 and P_2 for an ideal gas are shown in figure. Which is correct?



- (1) $P_1 > P_2$
(2) $P_1 < P_2$
(3) $P_1 = P_2$
(4) All of these
69. 60 ml of SO_2 effuses through a vessel in 10 sec. X ml of CH_4 effuses out in 10 sec under similar condition. find X?
- (1) 60 (2) 120 (3) 240 (4) 30
70. An oxide of sulphur contains 50% S. What will be empirical formula?
- (1) SO (2) SO_2
(3) SO_3 (4) S_2O_3
71. In the reaction :-

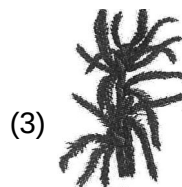
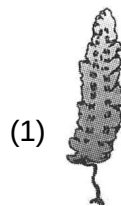
$$2\text{Al}_{(s)} + 6\text{HCl}_{(aq)} \rightarrow 2\text{Al}^{3+}_{(aq)} + 6\text{Cl}^{-}_{(aq)} + 3\text{H}_{2(g)}$$
 (1) 6L $\text{HCl}_{(aq)}$ is consumed for every 3L $\text{H}_{2(g)}$ produced
 (2) 33.6 L $\text{H}_{2(g)}$ is produced regardless of temperature and pressure for every mole Al that reacts
 (3) 67.2 L $\text{H}_{2(g)}$ at STP is produced for every mole Al that react
 (4) 11.2 L $\text{H}_{2(g)}$ at STP is produced for every mole $\text{HCl}_{(aq)}$ consumed
72. Oxidation number of Fe in $\text{Fe}_{0.94}\text{O}$ is :-
 (1) 200 (2) 200/94
 (3) 94/200 (4) none of these
73. When KMnO_4 is reduced with oxalic acid in acid medium, the oxidation number of Mn changes from :-
 (1) +7 to +4 (2) +6 to +4
 (3) +7 to +2 (4) +4 to +2

74. Number of moles present in 180 g of glucose :-
 (1) 1 (2) 4 (3) 5 (4) 6
75. The shape of $[\text{ClF}_4]^-$ and $[\text{ClF}_2]^-$ ions is respectively :-
 (1) See-saw and linear.
 (2) See-saw and bent.
 (3) Tetrahedral and linear.
 (4) Square planar and linear.
76. The hybridization of all carbon atoms in benzene is :-
 (1) sp^2
 (2) sp^3
 (3) sp^2 and sp^3
 (4) sp^2 and sp
77. Which of the following statements is correct?
 (1) Noble gases are insoluble in water.
 (2) The solubility of noble gases in water is due to London dispersion force.
 (3) The solubility of noble gases increases with the decrease in size of the noble gas atom.
 (4) The solubility of noble gases in water is due to dipole-induced dipole interaction.
78. Which of the following molecules has the maximum number of A-X bonds of identical length, where 'A' is the central atom and 'X' is the surrounding atom?
 (1) SF_6 (2) IF_7
 (3) PF_5 (4) ClO_4^-
79. Which of the following is the most soluble in water?
 (1) CsClO_4
 (2) NaClO_4
 (3) KClO_4
 (4) LiClO_4
80. The most abundant element in the Universe is :-
 (1) hydrogen
 (2) oxygen
 (3) nitrogen
 (4) helium

- | | |
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| <p>81. The phytohormone which can replace low temperature requirement for flowering in some plant is</p> <p>(1) Vernalin
(2) Ethylene
(3) Cytokinin
(4) Gibberellin</p> <p>82. For flowering critical dark period should always be exceeded in</p> <p>(1) Long day plant
(2) Short day plant
(3) Day neutral plant
(4) All type of plants</p> <p>83. Chilling treatment prior to flowering</p> <p>(1) Induce early flowering
(2) Is essential for biennials
(3) Inhibit reproductive growth
(4) More than one option is correct</p> <p>84. In which of the following steps during respiration CO_2 is not released?</p> <p>(1) Glycolysis
(2) Lactic acid fermentation
(3) Oxidation of isocitrate to oxalosuccinate
(4) All of these</p> <p>85. Cytochrome b and Cyt c_1 is a part of complex</p> <p>(1) Complex I
(2) Complex III
(3) Complex V
(4) Complex II</p> <p>86. One glucose on complete oxidation yield 36 ATP. During this oxidation of glucose what is the contribution of Kreb's cycle?</p> <p>(1) 24 ATP
(2) 12 ATP
(3) 15 ATP
(4) 30 ATP</p> | <p>87. All the enzymes of Kreb's cycle are present in mitochondrial matrix, except</p> <p>(1) Succinate dehydrogenase
(2) Isocitrate dehydrogenase
(3) Succinate thiokinase
(4) Aconitase</p> <p>88. A five carbon compound of Kreb's cycle which undergoes oxidative decarboxylation is</p> <p>(1) Pyruvic acid
(2) Isocitrate
(3) α-Ketoglutarate
(4) Oxalosuccinate</p> <p>89. The enzyme of ETS which has two copper centres associated with it is</p> <p>(1) Cytochrome reductase
(2) Cytochrome oxidase
(3) Succinate dehydrogenase
(4) FADH dehydrogenase</p> <p>90. The movement of pyruvate into mitochondria is a type of</p> <p>(1) Active transport
(2) Simple diffusion
(3) Co-Transport
(4) Facilitated diffusion</p> <p>91. Optimum CO_2 concentration for sugarcane and wheat are:</p> <p>(1) $360 \mu\text{L L}^{-1}$ and $450 \mu\text{L L}^{-1}$
(2) $450 \mu\text{L L}^{-1}$ and $360 \mu\text{L L}^{-1}$
(3) $360 \mu\text{L L}^{-1}$ and $360 \mu\text{L L}^{-1}$
(4) $450 \mu\text{L L}^{-1}$ and $450 \mu\text{L L}^{-1}$</p> <p>92. Compensation point refers to</p> <p>(1) Rate of photosynthesis
(2) Beginning of photosynthesis
(3) Rate of photosynthesis being equal to the rate of respiration
(4) Rate of photorespiration being equal to the rate of cellular respiration</p> |
|--|--|

93. The rate of photosynthesis per unit of light energy would be more in
- (1) Continuous bright light
 - (2) Continuous diffuse light
 - (3) Intermittent light
 - (4) Normal green light
94. Calvin cycle operates in
- (1) All photosynthetic plants
 - (2) C_3 plants only
 - (3) Most of C_3 plants and some C_4 plants
 - (4) C_3 plants and CAM plants
95. Stomatal opening is induced by
- (1) High CO_2 concentration, low pH, light
 - (2) Low CO_2 concentration, low pH, ABA
 - (3) Low CO_2 concentration, high pH, light
 - (4) Light, high temperature, ABA
96. Stages in proper sequence of prophase I are
- (1) Leptotene, Zygotene, Pachytene, Diplotene, Diakinesis
 - (2) Leptotene, Pachytene, Diplotene, Zygotene, Diakinesis
 - (3) Leptotene, Diplotene, Diakinesis, Pachytene, Zygotene
 - (4) Leptotene, Zygotene, Pachytene, Diakinesis, Diplotene
97. Pachytene stage is characterised by
- (1) Formation of synaptonemal complex
 - (2) Exchange of genetic material between nonsister chromatid of homologous chromosome
 - (3) Condensation of chromatin material
 - (4) Formation of monovalents

98. Polysome is formed by
- (1) A ribosome with several subunits
 - (2) Ribosomes attached to each other in a linear arrangement
 - (3) Several ribosomes attached to a single mRNA
 - (4) Many ribosomes attached to a strand of endoplasmic reticulum
99. The main arena of various types of activities of a cell is
- (1) Plasma membrane
 - (2) Mitochondrion
 - (3) Cytoplasm
 - (4) Nucleus
100. A plant which provides peat that have long been used as fuel and because of their capacity to hold water as packing material for transshipment of living material is



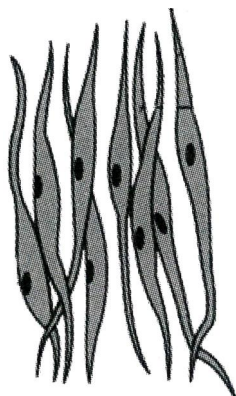
101. The term histology, i.e. study, of tissue was given by

- (1) Malpighi (2) Mayer
(3) Harvey (4) Bichat

102. Cells of germinal epithelium are

- (1) Columnar (2) Ciliated
(3) Squamous (4) Cuboidal

103. The following muscles are located in



- (1) Heart (2) Limbs
(3) Tongue (4) Wall of intestine

104. Formation of antibodies is the function of

- (1) Neutrophils (2) Lymphocytes
(3) Basophils (4) Mast cells

105. Ossification is the process by which

- (1) Bone is formed
(2) Wound healing takes place
(3) Cartilage is formed
(4) Skin is formed

106. Blood is a

- (1) Dense irregular tissue
(2) Muscular tissue
(3) Connective tissue
(4) Epithelial tissue

107. Nervous tissue originates from

- (1) Ectoderm
(2) Extra embryonic mesoderm
(3) Mesoderm and endoderm
(4) Endoderm

108. Centrosomes are absent in

- (1) Epithelial cells
(2) Germinal cells
(3) Nerve cells
(4) Muscle cells

109. Number of segments in thorax of cockroach is

- (1) Two (2) Three
(3) Four (4) Ten

110. Human dentition is

- (1) Diphyodont (2) Thecodont
(3) Heterodont (4) All of these

111. Mineral used for blood clotting is

- (1) Sodium (2) Calcium
(3) Nickel (4) Zinc

112. Voice box of birds is called

- (1) Furcula (2) Syrinx
(3) Synsacrum (4) Keel

113. Which of the following structures are present in mediastinum?

- (1) Heart (2) Oesophagus
(3) Thymus (4) All of these

114. _____ is called the graveyard of RBCs.

- (1) Liver (2) Spleen
(3) Kidney (4) Heart

115. If heart is shifted to right side, the condition is called

- (1) Dextrocardia (2) Mesocardia
(3) Oedema (4) Heart block

116. Longest loop of Henle are found in

- (1) Frog (2) Horse
(3) Camel (4) Kangaroo rat

117. The osmolality of interstitial fluid in cortex of Kidney is

- (1) 1200 mOsm/L
(2) 300 mOsm/L
(3) 900 mOsm/L
(4) 600 mOsm/L

118. Dorsal root ganglia of spinal cord have

- (1) Pseudounipolar (2) Unipolar
(3) Multipolar (4) Apolar

119. In resting state of neuron :-

- (1) There are more Na^+ outside
(2) There are more K^+ inside
(3) Negatively charged proteins are trapped inside
(4) All are correct

120. Hormones are

- (1) Chemical messengers
(2) Produced by endocrinal glands
(3) Diverse in their chemical nature
(4) All options are correct

Direction for Assertion & Reason questions. These questions consist of two statements each, printed as assertion and reason. While answering these questions you are required to choose any one of the following four responses.

- (1) If both Assertion & Reason are true & the reason is a correct explanation of the Assertion.
- (2) If both Assertion & Reason are true but reason is not a correct explanation of the Assertion.
- (3) If Assertion is true but the Reason is false
- (4) If both Assertion & Reason are false

121. *Assertion* : Equal masses of helium and oxygen gases are given equal quantities of heat. There will be a greater rise in the temperature of helium compared to that of oxygen.

Reason : The molecular weight of oxygen is more than the molecular weight of helium.

122. *Assertion* : Cream gets separated out of milk when it is churned, it is due to gravitational force.

Reason : In circular motion gravitational force is equal to centripetal force.

123. *Assertion* : The passengers sitting in a bus fall backward, when the bus suddenly starts moving.

Reason : Every body has the inability to change by itself, its state of rest.

124. *Assertion* : The ratio of specific heat gas at constant pressure and specific heat at constant volume for a diatomic gas is more than that for a monatomic gas.

Reason : The molecules of a monatomic gas have more degree of freedom than those of a diatomic gas.

125. *Assertion* : Pulling a lawn roller is easier than pushing it.

Reason : Pushing increases the apparent weight and hence friction

126. *Assertion* : The internal energy of a real gas is function of both, temperature and volume.

Reason : Internal kinetic energy depends on temperature and internal potential energy depends on volume.

127. *Assertion* : A man who falls from a height on a cement floor receive more injury than when he falls from the same height on a heap of sand.

Reason : The impulse given by cement floor is more than the impulse given by a heap of sand.

128. *Assertion* : The equation of motion can be applied only if the acceleration is along the direction of velocity and is constant.

Reason : If the acceleration of a body is constant then its motion is known as uniform motion.

129. *Assertion* : Two balls of different masses are thrown vertically upward with same speed. They will pass through their point of projection in the downward direction with the same speed.

Reason : The maximum height and downward velocity attained at the point of projection are independent of the mass of the ball.

130. *Assertion* : Velocity-time graph of an object in uniform motion along a straight path is a straight line parallel to the time axis.

Reason : In uniform motion of an object, velocity increases as the square of time elapsed.

131. *Assertion* : For an ideal gas, at constant temperature, the product of the pressure and volume is constant.

Reason : The mean square velocity of the molecules is inversely proportional to mass.

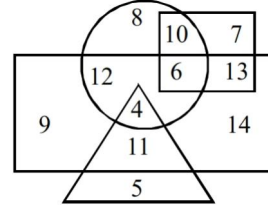
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| <p>132. <i>Assertion</i> : For the body to be in rotational equilibrium the net torque acting on the body about any point is zero.
 <i>Reason</i> : For net torque to be zero, net force should also be zero.</p> <p>133. <i>Assertion</i> : Animals curl into a ball, when they feel very cold.
 <i>Reason</i> : Animals by curling their body reduces the surface area.</p> <p>134. <i>Assertion</i> : The average and instantaneous velocities have same value in a uniform motion.
 <i>Reason</i> : In uniform motion, the velocity of an object increases uniformly.</p> <p>135. <i>Assertion</i> : Static friction is self adjusting force
 <i>Reason</i> : The magnitude of static friction is less than the applied force.</p> <p>136. <i>Assertion</i> : Position of the centre of mass can lie outside the body also.
 <i>Reason</i> : Centre of mass doesn't shift its position in the absence of an external force.</p> <p>137. <i>Assertion</i> : When a rigid body rotates about any axis, then all the particles of it move in circles of different radii but with same angular velocity.
 <i>Reason</i> : In rigid body relative position of particles are fixed.</p> <p>138. <i>Assertion</i> : When a gas is allowed to expand, work done by gas is positive.
 <i>Reason</i> : Force due to gaseous pressure and displacement (of piston) are in the same direction.</p> <p>139. <i>Assertion</i> : Two thin blankets put together are warmer than a single blanket of double the thickness.
 <i>Reason</i> : Thickness increases because of air layer enclosed between the two blankets.</p> <p>140. <i>Assertion</i> : Radiation is the speediest mode of heat transfer.
 <i>Reason</i> : Radiation can be transmitted in zig-zag motion.</p> | <p>141. <i>Assertion</i> : A reaction which is spontaneous and accompanied by decrease randomness must be exothermic.
 <i>Reason</i> : All exothermic reactions are accompanied by decreases of randomness.</p> <p>142. <i>Assertion</i> : Combustion of N_2 to given NO is endothermic.
 <i>Reason</i> : $N \equiv N$ bond is very strong and so has high bond energy.</p> <p>143. <i>Assertion</i> : The pH of pure water is less than 7 at $60^\circ C$.
 <i>Reason</i> : As the temperature increases, pure water becomes slightly acidic.</p> <p>144. <i>Assertion</i> : 0.1 M NaCN + 0.05 M HCl solution on mixing in equal volume forms a buffer solution.
 <i>Reason</i> : The solution is a mixture of salt and acid, thus acts as a buffer.</p> <p>145. <i>Assertion</i> : The pressure of real gases is less than the pressure of the ideal gas.
 <i>Reason</i> : The intermolecular forces of attraction in real gases are greater than those of ideal gas.</p> <p>146. <i>Assertion</i> : The value of van der Waal's constant 'a' is higher for ammonia than for nitrogen.
 <i>Reason</i> : Intermolecular hydrogen bonding is present in ammonia.</p> <p>147. <i>Assertion</i> : Different gases at the same conditions of temperature and pressure have same root mean square speed.
 <i>Reason</i> : Average kinetic energy of a gas is directly proportional to temperature in degree centigrade scale.</p> <p>148. <i>Assertion</i> : 22.4 L of He gas is having N_A atoms of He at STP.
 <i>Reason</i> : 22.4 L of any gas at STP is having N_A atoms.</p> <p>149. <i>Assertion</i> : Law of definite proportion is not applicable for the compounds of Fe or Cu.
 <i>Reason</i> : Law of definite proportion is not applicable for those metal which shows variable valency.</p> <p>150. <i>Assertion</i> : Among $KMnO_4$, $KClO_4$, HNO_3 only HNO_3 acts as oxidant while rest two may also act as reductant.
 <i>Reason</i> : In these molecules central atom is in its lowest oxidation state.</p> |
|---|---|

- | | |
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| <p>151. <i>Assertion</i> : In H_2SO_5, oxidation number of S is +6.
 <i>Reason</i> : There is one peroxide linkage.</p> <p>152. <i>Assertion</i> : The bond order decreases by 0.5 when N_2 converts to NO_2^+.
 <i>Reason</i> : The removal of an electron takes place from π_{bonding} orbital when N_2 converts to NO_2^+.</p> <p>153. <i>Assertion</i> : Mercurous ion is always diamagnetic.
 <i>Reason</i> : The 79th electron of each Hg atom gets paired up to form a bond between two Hg atoms.</p> <p>154. <i>Assertion</i> : The $d_{\text{O-O}}$ in O_2F_2 is almost same with that in O_2 molecule.
 <i>Reason</i> : O_2F_2 exists in the form of $[\text{O}=\text{O}^+-\text{F}]^-$ also.</p> <p>155. <i>Assertion</i> : CaCl_2 is formed as a by-product in the Solvay process for Na_2CO_3 preparation.
 <i>Reason</i> : During the recovery of NH_3, CaCl_2 is obtained when $\text{Ca}(\text{OH})_2$ is used to react with NH_4Cl.</p> <p>156. <i>Assertion</i> : Silicones have water repelling characteristics.
 <i>Reason</i> : $(\text{Si}-\text{O}-\text{Si})$ skeleton is covered with alkyl groups.</p> <p>157. <i>Assertion</i> : Free protons (H^+) do not exist under normal conditions.
 <i>Reason</i> : H^+ ions is extremely small in size and has a very polarizing power.</p> <p>158. <i>Assertion</i> : Neopentane forms only one monosubstituted compound.
 <i>Reason</i> : It has only one type of carbon atoms.</p> <p>159. <i>Assertion</i> : Ethanol can show position isomerism.
 <i>Reason</i> : Ethanol cannot show isomerism.</p> <p>160. <i>Assertion</i> : Phenol is less acidic than p-nitrophenol
 <i>Reason</i> : Phenolate ion is more stable than p-nitro phenolate ion.</p> | <p>161. <i>Assertion</i> : Heterophylly is an example of plasticity.
 <i>Reason</i> : Heterophylly means different forms of leaves on the same plant.</p> <p>162. <i>Assertion</i> : Auxins are widely used as herbicides.
 <i>Reason</i> : 2,4 D is used to kill monocotyledonous weeds.</p> <p>163. <i>Assertion</i> : Presence of apical bud inhibits the growth of the lateral buds.
 <i>Reason</i> : Gibberellin are mainly responsible for apical dominance.</p> <p>164. <i>Assertion</i> : Glycolysis is a common pathway for both aerobic and anaerobic respiration.
 <i>Reason</i> : Glycolysis is one of the ancient metabolic pathway in living cells.</p> <p>165. <i>Assertion</i> : PFK is rate limiting enzyme of glycolysis.
 <i>Reason</i> : It is regarded as pacemaker enzyme of glycolysis.</p> <p>166. <i>Assertion</i> : Photorespiration protects plants from photo oxidation.
 <i>Reason</i> : Light induced reactions may damage photosynthetic activity.</p> <p>167. <i>Assertion</i> : If a cell is not to divide it will enter into G_0 phase.
 <i>Reason</i> : Cells of G_0 phase are metabolically inactive and non proliferating.</p> <p>168. <i>Assertion</i> : In plasma membrane extrinsic proteins are abundant towards cytosolic face.
 <i>Reason</i> : Proteins are arranged asymmetrically and shows flip flop movement.</p> <p>169. <i>Assertion</i> : According to five kingdom system of classification <i>Chlamydomonas</i> and <i>Chlorella</i> are placed in kingdom Protista.
 <i>Reason</i> : Both are unicellular eukaryotic organisms.</p> <p>170. <i>Assertion</i> : Deuteromycetes is commonly known as Fungi imperfecti.
 <i>Reason</i> : They lack perfect or sexual stage.</p> |
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| <p>171. <i>Assertion</i> : RBCs of mammals are enucleated.
 <i>Reason</i> : Absence of nucleus helps to accommodate maximum amount of haemoglobin.</p> <p>172. <i>Assertion</i> : It is beneficial to breathe through nose that to breathe through mouth.
 <i>Reason</i> : Nasal chambers cause sterilization and humidification of air.</p> <p>173. <i>Assertion</i> : Heart is enclosed within two layered covering called pericardium.
 <i>Reason</i> : Cardiac muscles are present in myocardium.</p> <p>174. <i>Assertion</i> : Human heart is myogenic.
 <i>Reason</i> : The impulse for each heart beat is generated within the heart in a muscular node called pacemaker (SA node).</p> <p>175. <i>Assertion</i> : Patients with diabetes insipidus have polyuria and glycosuria.
 <i>Reason</i> : Vasopressin is synthesized in neurohypophysis.</p> <p>176. <i>Assertion</i> : Human skull is dicondylic.
 <i>Reason</i> : Atlas has two condyles using which it rests on vertebral column.</p> <p>177. <i>Assertion</i> : In neurons, the action potential is generated by influx of sodium ions.
 <i>Reason</i> : Efflux of potassium ions cause depolarization</p> <p>178. <i>Assertion</i> : Mammals and aves are homeothermic animals.
 <i>Reason</i> : Mammals and aves can maintain a constant body temperature despite variations of temperature in surroundings.</p> <p>179. <i>Assertion</i> : Reptiles, aves and mammals are called amniotes.
 <i>Reason</i> : Reptiles, aves and mammals have 12 pairs of cranial nerves and metanephric kidneys.</p> <p>180. <i>Assertion</i> : Retrogressive metamorphosis is seen in urochordates.
 <i>Reason</i> : In urochordates notochord is absent in larval stage, but is present in tail of adult.</p> | <p>181. Who has won the best actor award at the 90th Academy Awards 2018?
 (1) Matt Damon
 (2) Gary Oldman
 (3) Denzel Washington
 (4) Leonardo DiCaprio</p> <p>182. Who is the first Indian cricketer to cross 900 points in the ICC ODI player rankings ?
 (1) Ajinkya Rahane
 (2) Ravichandran Ashwin
 (3) Hardik Pandya
 (4) Virat Kohli</p> <p>183. What is the India's rank in the world in terms of military power, as per Global Firepower Index -2017?
 (1) 8th (2) 6th
 (3) 4th (4) 5th</p> <p>184. Stephen Hawking, the legendary theoretical physicist belonged to which country?
 (1) United States America
 (2) France
 (3) United Kingdom
 (4) Germany</p> <p>185. What is the name of India's first online Radio Station?
 (1) Radio Talk
 (2) Radio Zingle
 (3) Radio Garden
 (4) Radio Umang</p> <p>186. Which is the book written by PM Narendra Modi for students?
 (1) Aahwan
 (2) A Journey
 (3) You Can Win
 (4) Exam Warriors</p> |
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187. Which one is the first state to have its own flag?
- (1) Karnataka
 - (2) Maharashtra
 - (3) Jammu and Kashmir
 - (4) Tamil Nadu
188. World Consumer Rights Day is observed on which date every year?
- (1) March 14
 - (2) March 15
 - (3) March 16
 - (4) March 17
189. After the Bardoli Satyagraha, the title of 'Sardar' to Vallabhbhai Patel was given by
- (1) Jawaharlal Nehru
 - (2) Motilal Nehru
 - (3) Mahatma Gandhi
 - (4) Maulana Abul Kalam Azad
190. Who of the following is/was the oldest to take the oath as the Prime Minister of India?
- (1) Atal Bihari Vajpayee
 - (2) Manmohan Singh
 - (3) Morarji Desai
 - (4) PV Narasimha Rao
191. Find out the wrong number in the series.
- 196, 169, 144, 121, 100, 80, 64
- (1) 169
 - (2) 144
 - (3) 121
 - (4) 80
192. Running at the same constant rate, 6 identical machines can produce a total of 270 bottles per minute. At this rate, how many bottles could 10 such machines produce in 4 minutes?
- (1) 648
 - (2) 1800
 - (3) 2700
 - (4) 10800

193. In the following diagram Rectangle represents men, Triangle represents educated, Circle represents urban and Square represents government employees.

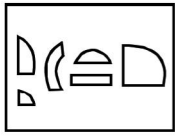


Which one of the following represents the educated men but not urban ?

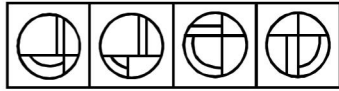
- (1) 9
 - (2) 5
 - (3) 4
 - (4) 11
194. Choose the word which is different from the rest.
- (1) Cap
 - (2) Turban
 - (3) Hat
 - (4) Veil
195. A man has some hens and cows. If the number of heads be 48 and the number of feet equals 140, then the number of hens will be
- (1) 22
 - (2) 23
 - (3) 24
 - (4) 26
196. A library has an average of 510 visitors on Sundays and 240 on other days. The average number of visitors per day in a month of 30 days beginning with a Sunday is :-

- (1) 250
- (2) 276
- (3) 280
- (4) 285

197. Find out which of the figures (1), (2), (3) and (4) can be formed from the pieces given in figure (X).



(X)



(1)

(2)

(3)

(4)

(1) 1

(2) 2

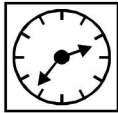
(3) 3

(4) 4

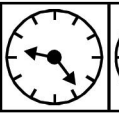
198. Choose the correct mirror image of the given figure (X) from amongst the four alternatives.



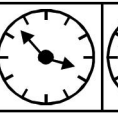
(X)



(1)



(2)



(3)



(4)

199. Rakesh is travelling on his cycle and has calculated to reach point A at 2 P.M. if he travels at 10 kmph, he will reach there at 12 noon if he travels at 15 kmph. At what speed must he travel to reach A at 1 P.M.?

(1) 8 kmph

(2) 11 kmph

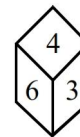
(3) 12 kmph

(4) 14 kmph

200. Three different positions X, Y and Z of a dice are shown in the figures given below. Which numbers are hidden behind the numbers 6 and 5 in the position Z?



(X)



(Y)



(Z)

(1) 1 and 4

(2) 1 and 3

(3) 4 and 3

(4) 1 and 2